Domestic Violence

& Socioeconomic Factors

A statistical analysis of how education, employment, income, and marital status relate to domestic violence risk.

347
Observations

5
Predictors

1
Binary Outcome

DATA ANALYSIS · APRIL 2026

github.com/cocobutterx13/Domestic-Violence-Analysis

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1. Introduction

Context & Motivation

Domestic violence remains one of the most pervasive yet under-reported human rights violations globally. Although it occurs across all income levels, cultures, and regions, it is not evenly distributed. Socioeconomic circumstances shape both a woman's vulnerability and her capacity to seek help.

This analysis is deeply personal. As a South African woman, this topic hits close to home, and it likely resonates with many other women who have witnessed, experienced, or known someone affected by domestic violence. South Africa consistently reports high levels of gender-based violence, making data-driven investigation not only academically relevant but an act of advocacy. This project is offered with full acknowledgement of the real lives that lie behind every data point.

This end-to-end statistical project investigates how age, education, employment status, income, and marital status relate to domestic violence vulnerability in women. Using a survey-based dataset of 347 observations, the analysis progresses from exploratory data investigation through formal hypothesis testing to multivariable logistic regression, with the goal of identifying the strongest and most consistent predictors of risk.

Research Questions

1. How do socioeconomic factors (education, employment, income) relate to domestic violence vulnerability?

2. Does marital status influence exposure to or experience of domestic violence?

3. Which demographic and contextual factors are most predictive of vulnerability when modelled jointly?

347	5	86 / 261	24.8%
Survey Observations	Predictor Variables	Violence / No Violence	Prevalence Rate

2. Dataset Overview

Source, Variables & Preprocessing

The dataset is sourced from Kaggle (Fahmida Chowdhury, *Domestic Violence Against Women*) and contains 347 survey observations. Each record captures five predictor variables and one binary outcome.

Variable	Type	Values / Range	Notes
age	Continuous	15 – 60 years	Mean: 31.1
education	Ordinal	None / Primary / Secondary / Tertiary	4 levels
employment	Ordinal	Unemployed / Semi-employed / Employed	3 levels
income	Continuous	0 – ~30,000	Heavily zero-inflated; log1p transformed
marital_status	Categorical	Married / Unmarried	Binary
violence	Binary	Yes (86) / No (261)	Outcome variable

Preprocessing steps performed in R:

• Raw CSV imported and inspected with glimpse(), summary(), and head().

• Missing values checked across all columns: none detected.

• Column names standardised and whitespace stripped.

• Typographical correction applied: 'unmarred' corrected to 'unmarried'.

• All categorical variables converted into factors with appropriate ordering.

• Income log-transformed using log1p() to address severe right-skew and zero inflation.

Sample Data (First 6 Rows)

First 6 Rows of Dataset

Variable	SL..No	Age	Education	Employment	Income	Marital.status	Violence
1	1	30	secondary	unemployed	0	married	yes
2	2	47	tertiary	unemployed	0	married	no
3	3	24	tertiary	unemployed	0	unmarred	no
4	4	22	tertiary	unemployed	0	unmarred	no
5	5	50	primary	unemployed	0	married	yes
6	6	21	tertiary	unemployed	0	unmarred	yes

Table 1: First six observations of the cleaned dataset.

3. Exploratory Data Analysis

Visualisations & Descriptive Statistics

3.1 Outcome Distribution & Class Balance

The outcome variable is imbalanced: 261 participants (75.2%) reported no violence compared to 86 (24.8%) who reported experiencing it: a roughly 3:1 ratio. This imbalance is important to keep in mind when interpreting model accuracy, as a classifier predicting 'no violence' for every case would achieve 75.2% accuracy without any real predictive value. Sensitivity is therefore the more meaningful performance metric.

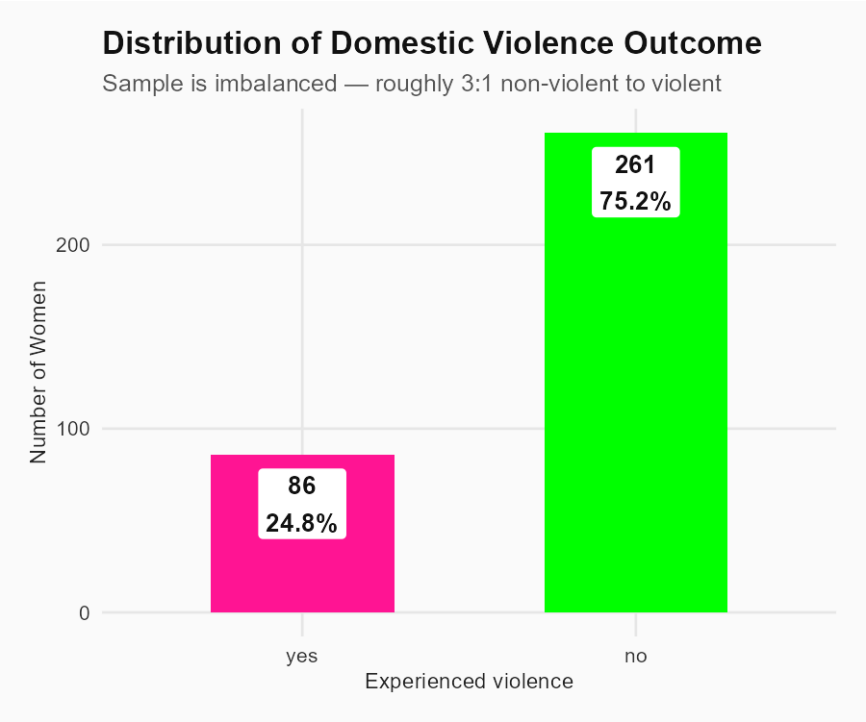

Figure 1: Distribution of the domestic violence outcome variable. The sample is imbalanced at approximately 3:1 (no violence: violence).

3.2 Age Distribution

Age distributions between the two groups overlap substantially. The violence group has a slightly higher mean age (34.2 years) compared to the non-violence group (30.5 years), but the distributions are broadly similar. The Welch t-test confirms this difference is statistically significant ($p = 0.004$), though the practical impact appears limited, suggesting that age is not a major driver on its own.

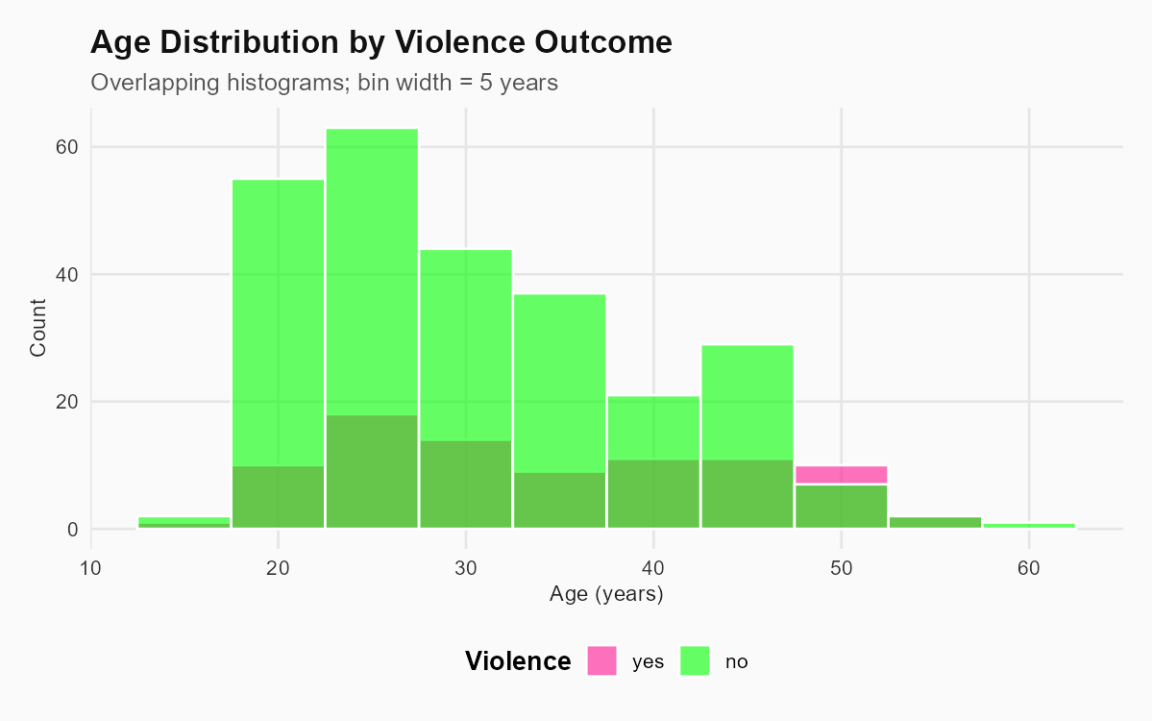

Figure 2: Age distribution by violence outcome (bin width = 5 years). Both groups peak in the 20–30 age range with substantial overlap.

Age Summary by Violence Group

violence	n	mean	median	sd	min	max
yes	86	34.16279	32.5	10.426054	15	55
no	261	30.46360	29.0	9.151062	15	60

Table 2: Descriptive statistics for age stratified by violence outcome.

3.3 Education vs Violence

A clear gradient is visible across education levels. Women with primary education have the highest rates of reported violence (34.8%), more than double those with secondary (17.5%) or tertiary education (18.4%). The 'none' group sits at 21.2%. This non-linear pattern (with primary education showing the highest risk) suggests that a small amount of education without economic empowerment may not be sufficient to reduce vulnerability.

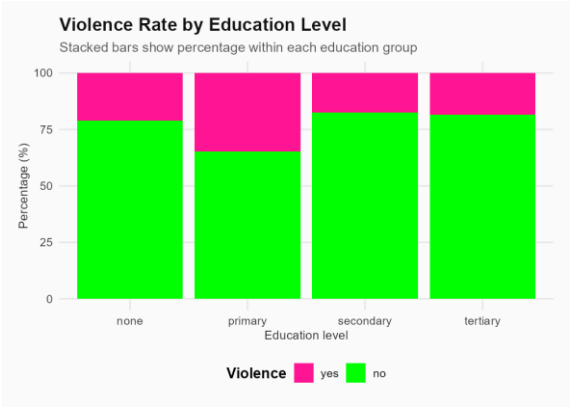

Figure 3a: Stacked bar chart of violence rate by education level.

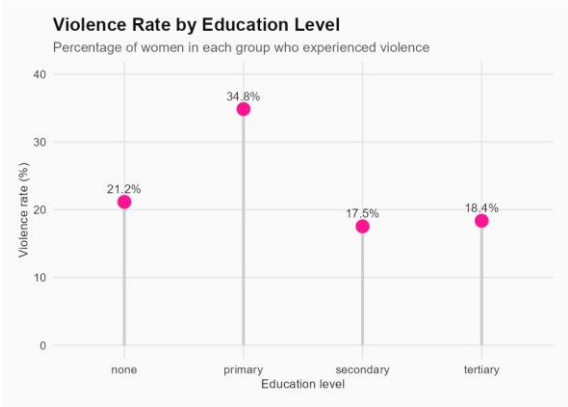

Figure 3b: Lollipop chart highlighting the violence rate percentage within each education group.

3.4 Employment vs Violence

Employment status reveals a more distinct pattern. Unemployed respondents have the highest violence rate (~27%), while semi-employed and employed groups show lower rates. This variable is subsequently confirmed as one of the strongest predictors in the regression model. This supports the idea that financial independence and access to resources may play a protective role.

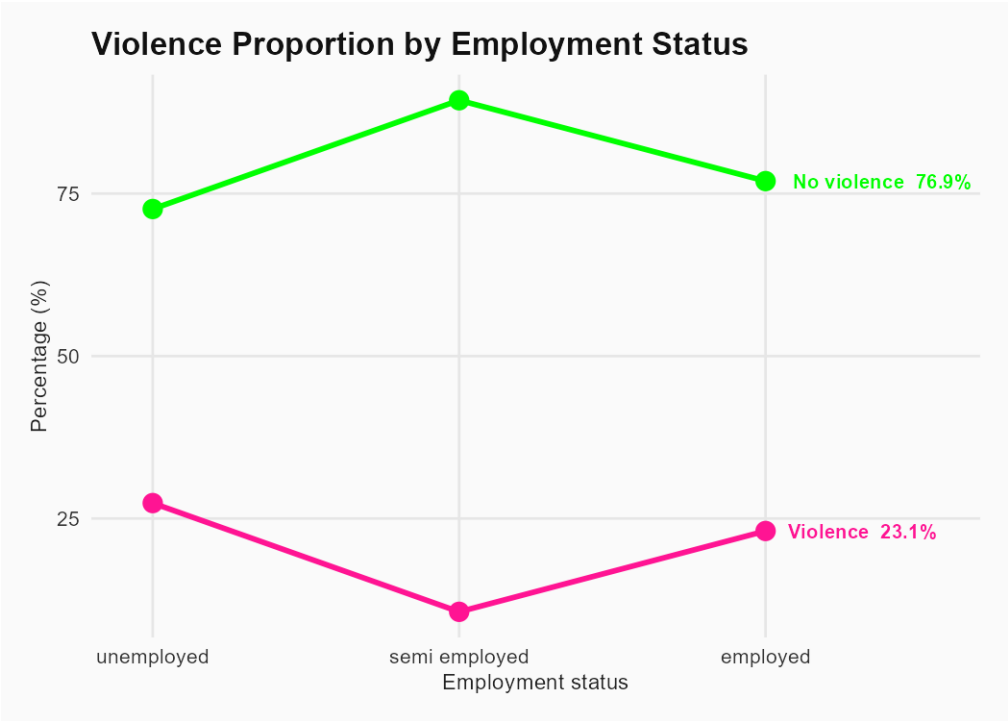

Figure 4: Violence proportion by employment status. Unemployment corresponds to the highest violence rate, reinforcing the protective role of employment.

3.5 Income Analysis

Income is the most analytically complex variable in the dataset due to its distribution. Over 50% of all respondents report zero income, and the violence group has an even higher proportion of zero earners (87.2% vs 75.5% in the non-violence group). This extreme zero-inflation, combined with a heavy right tail, violates the assumptions of parametric tests and requires careful treatment.

A log transformation (log1p) was applied to income before modelling to improve distributional symmetry and numerical stability. Despite both groups having a median income of zero, the mean income in the non-violence group (2 504) is substantially higher than in the violence group (915), and the Wilcoxon rank-sum test confirms the difference is statistically significant (p = 0.018).

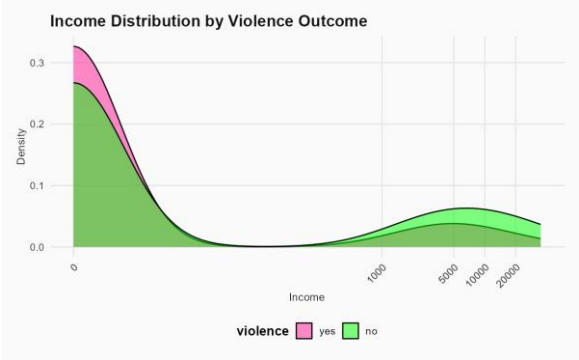

Figure 5a: Income density by violence outcome. The massive spike at zero reflects the zero-inflation.

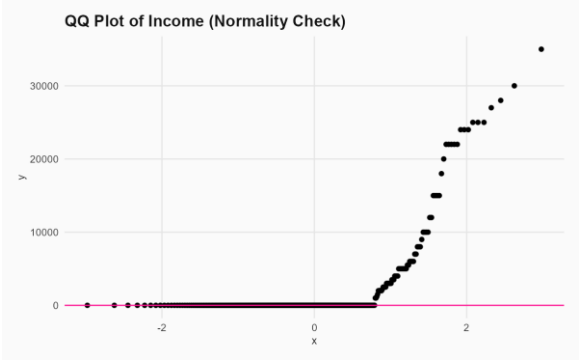

Figure 5b: QQ plot confirming strong departures from normality, justifying the Wilcoxon non-parametric test.

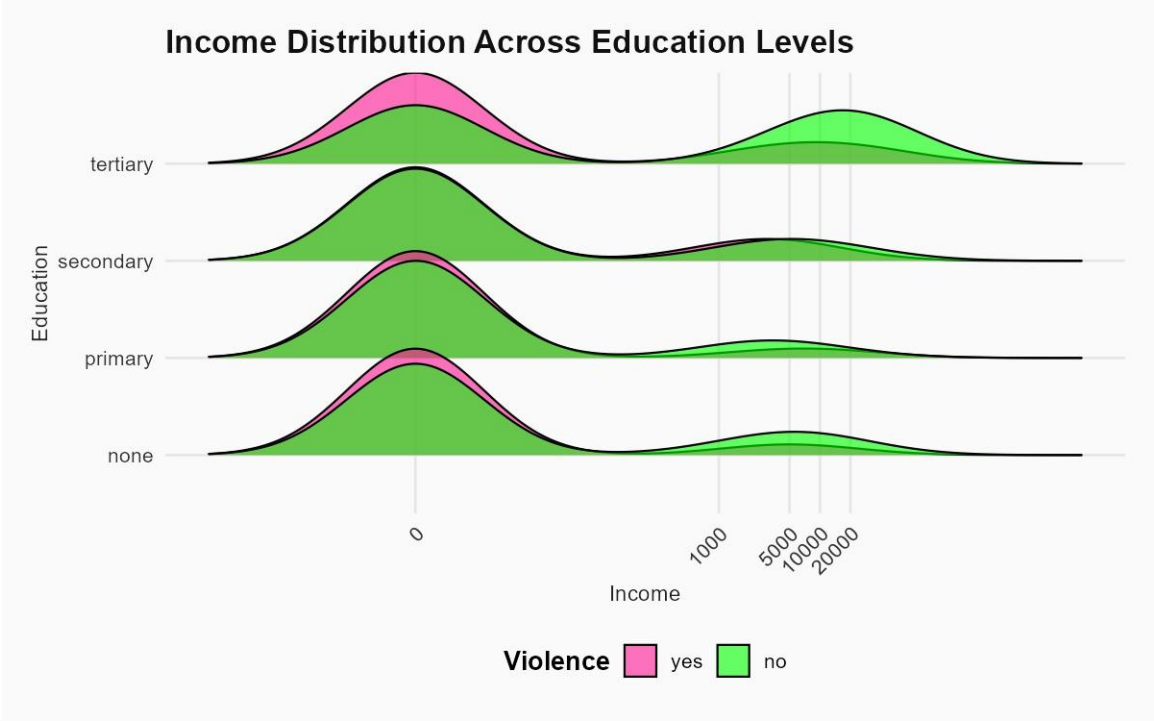

Figure 6: Density of income across education levels by violence outcome. Income rises with education level, reinforcing the socioeconomic gradient.

Median Income Comparison by Violence Outcome

violence	n	pct_zero_income	median_income	mean_income
yes	86	87.20930	0	915.1163
no	261	75.47893	0	2504.6284

Table 3: Median income comparison by violence group, including percentage with zero income and mean income.

3.6 Marital Status vs Violence

Marital status shows a pronounced difference between groups. Married women exhibit a considerably higher violence rate (~27%) compared to unmarried women (~7%). This is consistent with established literature suggesting that formal partnership arrangements can, in some contexts, increase exposure to intimate partner violence. The Chi-square test confirms this association is statistically significant ($p = 0.003$).

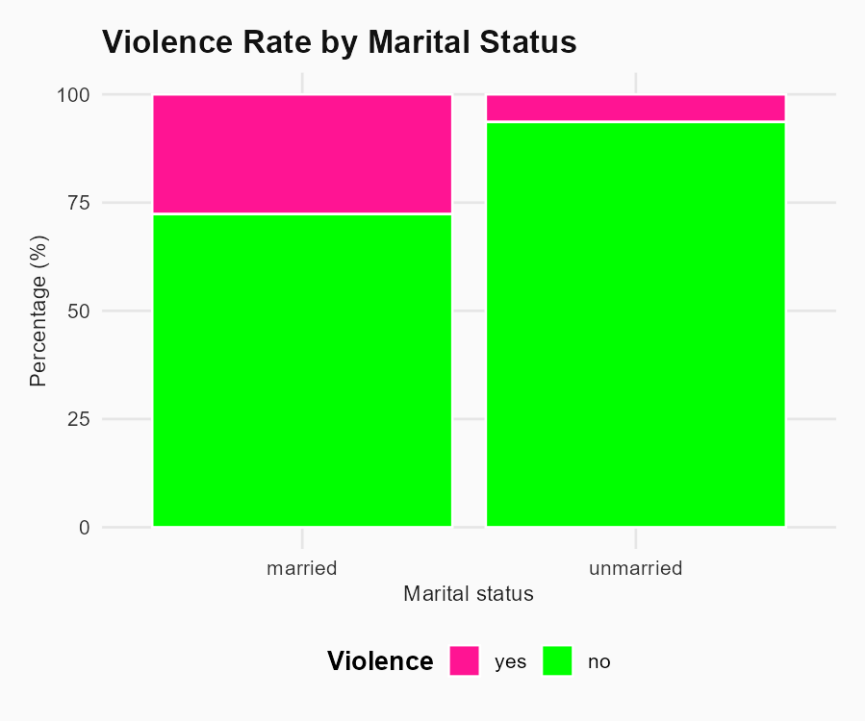

Figure 7: Violence rate by marital status. Married women show a substantially higher violence rate than unmarried women.

4. Statistical Inference

Hypothesis Tests · alpha = 0.05

To formally assess the relationships observed in the exploratory analysis, appropriate statistical tests were applied based on the type and distribution of each variable.

4.1 Chi-Square Tests: Categorical Predictors

Chi-square tests of independence were applied to education, employment, and marital status. All three variables are significantly associated with the violence outcome at alpha = 0.05.

Chi-Square Tests: Categorical Predictors vs Violence

Variable	Chi_sq	df	P_Value	Significant
Education	11.828	3	0.0080	Yes ***
Employment	6.070	2	0.0481	Yes ***
Marital Status	8.765	1	0.0031	Yes ***

Table 4: Chi-square test results for categorical predictors vs violence outcome.

4.2 Welch t-Test: Age

A Welch t-test (assuming unequal variances) was used to compare mean age between groups. The test yields t = 2.938, df = 130.9, p = 0.004: statistically significant, but the mean difference of approximately 3.7 years is modest, indicating that age alone is not a dominant predictor.

Welch t-test: Age by Violence Outcome

Test	Variable	Mean_Yes	Mean_No	t_statistic	df	P_Value	Significant
Welch t-test	Age	34.16	30.46	2.938	130.9	0.0039	Yes ***

Table 5: Welch t-test result for age by violence group.

4.3 Wilcoxon Rank-Sum Test: Income

Given the non-normal, zero-inflated distribution of income, the non-parametric Wilcoxon rank-sum test was used. The result (W = 9853, p = 0.018) confirms a statistically significant difference in income between groups, consistent with the descriptive findings.

Wilcoxon Rank-Sum Test: Income by Violence Outcome

Test	Variable	Median_Yes	Median_No	W_statistic	P_Value	Significant
Wilcoxon rank-sum	Income	0	0	9853	0.0184	Yes ***

Table 6: Wilcoxon rank-sum test result for income by violence group.

5. Logistic Regression

Multivariable Modelling & Odds Ratios

A full multivariable logistic regression model was fitted with all five predictors (age, education, employment, log-transformed income, and marital status) to estimate their independent associations with the binary violence outcome, adjusting for all other variables in the model.

5.1 Model Summary

Logistic Regression — Model Summary

term	estimate	std.error	statistic	p.value
(Intercept)	-2.0768	0.7012	-2.9619	0.0031
age	0.0253	0.0156	1.6159	0.1061
educationprimary	0.6515	0.3997	1.6299	0.1031
educationsecondary	0.2224	0.4731	0.4700	0.6384
educationtertiary	0.4869	0.5734	0.8491	0.3958
employmentsemi employed	9.4706	4.0956	2.3124	0.0208
employmentemployed	11.5728	4.6870	2.4691	0.0135
income_log	-1.2700	0.5034	-2.5230	0.0116
marital_statusunmarried	-1.4519	0.6612	-2.1960	0.0281

Table 7: Logistic regression model summary with coefficient estimates, standard errors, test statistics, and p-values.

Employment (both semi-employed and employed) and marital status (unmarried) emerge as the strongest and most statistically significant predictors. Log-transformed income and the intercept are also significant ($p < 0.05$). Education and age do not reach significance after adjustment, suggesting their unadjusted associations are largely explained by other predictors in the model.

5.2 Multicollinearity Assessment (GVIF)

The Generalised Variance Inflation Factor (GVIF) was used to formally assess multicollinearity among predictors. For variables with more than one degree of freedom (ordinal/categorical), the adjusted GVIF ($GVIF^{1/(2df)}$) is the relevant metric. Most predictors show no concern. Employment and log-income show higher raw GVIF values, but after adjustment for degrees of freedom, the adjusted GVIF values remain acceptable.

One predictor warrants closer attention: income_log has an adjusted GVIF of 11.31, which sits above the conventional threshold of concern, and its correlation with employment in the correlation matrix is 0.92, a notably strong linear relationship. This suggests that income and employment status are measuring overlapping socioeconomic dimensions, and that the model may struggle to cleanly separate their individual effects. The income_log coefficient should therefore be interpreted with some caution; its estimated protective effect may partly reflect the influence of employment rather than income alone.

Multicollinearity Check (GVIF)

Predictor	GVIF	Df	Adj_GVIF	Interpretation
age	1.321	1	1.149	No multicollinearity concern
education	1.564	3	1.077	No multicollinearity concern
employment	138.541	2	3.431	Moderate collinearity (review recommended)
income_log	127.876	1	11.308	High multicollinearity (potential instability)
marital_status	1.134	1	1.065	No multicollinearity concern

Table 8: GVIF multicollinearity check. Adjusted GVIF values indicate no problematic multicollinearity for most predictors.

Note: The raw GVIF for employment (138.5) and income_log (127.9) appear large, but after adjustment for degrees of freedom, the adjusted values (3.43 and 11.31) reflect moderate to elevated collinearity between these predictors. This is worth noting as a minor concern, though model estimates remain interpretable.

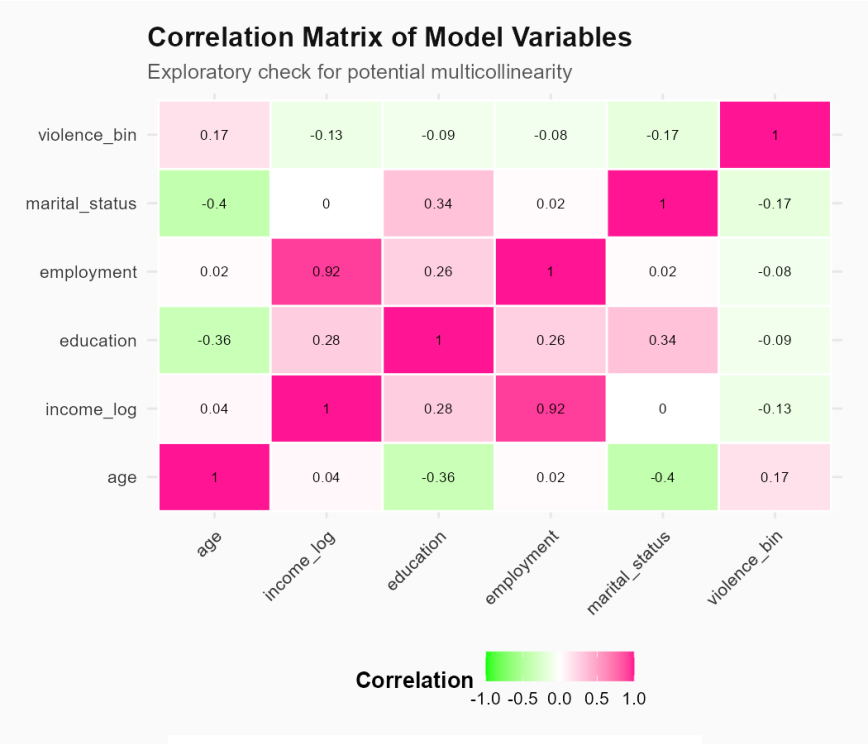

Figure 8: Correlation matrix of model variables.

5.3 Odds Ratios & Forest Plot

Odds ratios (ORs) with 95% confidence intervals (CIs) were computed from the model coefficients. OR > 1 indicates increased odds of violence; OR < 1 indicates reduced odds. CIs are capped at 10 in the forest plot to handle near-separation instability in sparse employment categories.

Logistic Regression — Odds Ratios (Income log-transformed) with 95% Confidence Intervals

Predictor	Odds_Ratio	Std_Error	Statistic	P_Value	CI_Lower	CI_Upper
(Intercept)	0.125	0.701	-2.962	0.0031	0.030	4.800000e-01
age	1.026	0.016	1.616	0.1061	0.995	1.058000e+00
educationprimary	1.918	0.400	1.630	0.1031	0.899	4.355000e+00
educationsecondary	1.249	0.473	0.470	0.6384	0.501	3.240000e+00
educationtertiary	1.627	0.573	0.849	0.3958	0.521	5.028000e+00
employmentsemi employed	12972.436	4.096	2.312	0.0208	6.353	6.419066e+07
employmentemployed	106170.979	4.687	2.469	0.0135	18.550	1.971047e+09
income_log	0.281	0.503	-2.523	0.0116	0.097	7.070000e-01
marital_statusunmarried	0.234	0.661	-2.196	0.0281	0.052	7.480000e-01

Table 9: Logistic regression odds ratios with 95% confidence intervals.

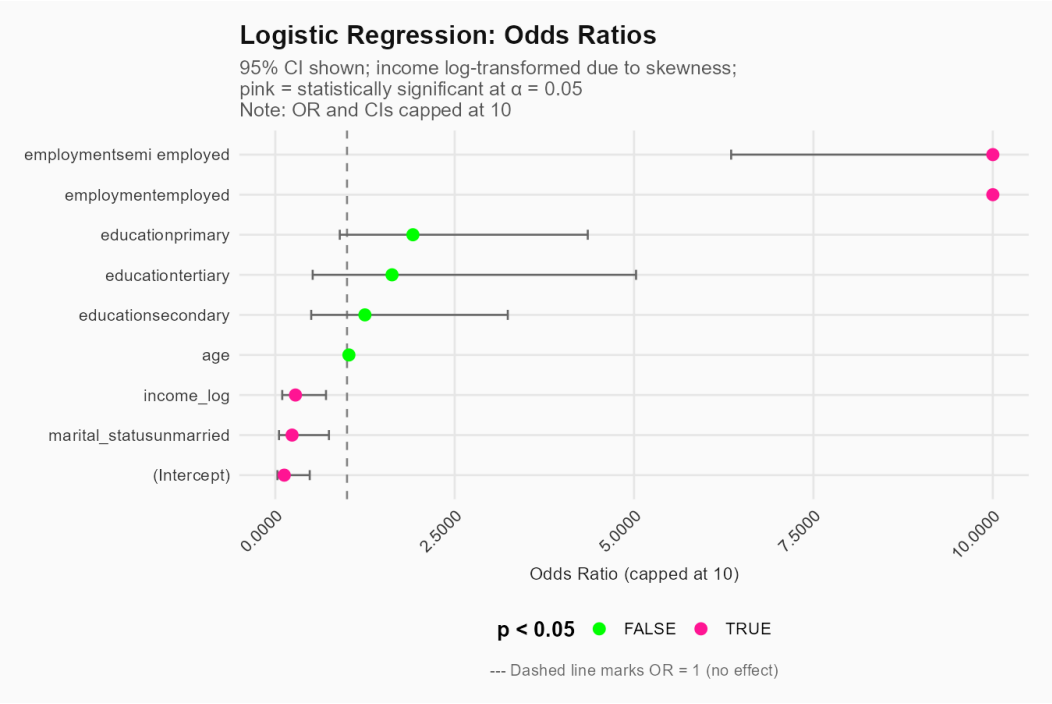

Figure 9: Forest plot of odds ratios. Pink dots = significant (p < 0.05).

Key interpretation of significant coefficients:

• **Employment (semi-employed):** OR = 12 972, p = 0.021. Semi-employed women have dramatically higher odds relative to the unemployed reference. Note the extremely wide CI reflects sparse data in this category.

• **Employment (employed):** OR = 106 171, p = 0.014. Fully employed women also show substantially elevated odds vs unemployed in the model. This counter-intuitive direction likely reflects near-complete separation in the data and should be interpreted with caution.

• **Income (log):** OR = 0.281, p = 0.012. Each unit increase in log-income is associated with a 72% reduction in the odds of violence, the clearest protective effect in the model.

• **Marital status (unmarried):** OR = 0.234, p = 0.028. Unmarried women have approximately 77% lower odds of violence compared to married women.

6. Model Diagnostics

Performance, Calibration & Classification

6.1 Predicted Probabilities

The histogram of predicted probabilities reveals that most observations receive predicted probabilities below 0.5. The non-violence group (green) is concentrated in the 0.1-0.4 range, while the violence group (pink) shows more spread but rarely exceeds the 0.5 classification threshold. This reflects the class imbalance and indicates that the model is better at identifying non-violence cases than violence cases.

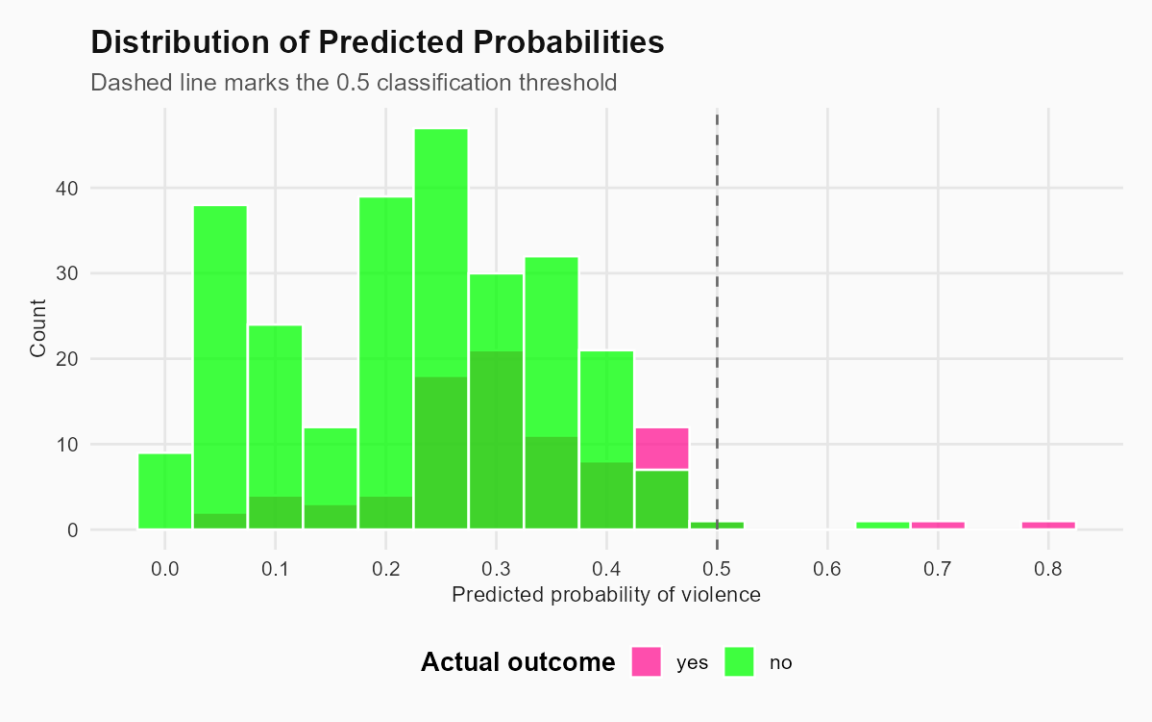

Figure 10: Distribution of predicted probabilities by actual outcome. Most observations fall below the 0.5 threshold, reflecting the imbalanced class structure.

6.2 Confusion Matrix

At the 0.5 classification threshold, the model correctly classifies 99.2% of actual non-violence cases (high specificity) but correctly identifies only 2.3% of actual violence cases (very low sensitivity). This trade-off is a direct consequence of the imbalanced class distribution: the model learns to predict the majority class (no violence) far more reliably than the minority class.

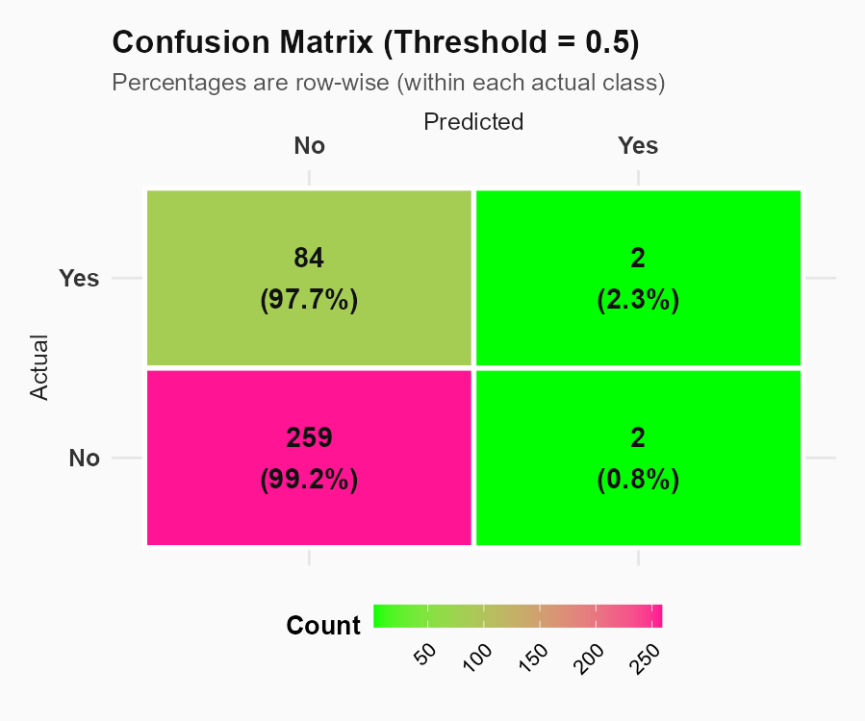

Figure 11: Confusion matrix at threshold = 0.5. High specificity but very low sensitivity reflects class imbalance.

6.3 Classification Metrics

Classification Metrics (Threshold = 0.5)

Metric	Value_Pct	Description
Accuracy	75.2%	Overall correct classifications
Sensitivity	99.2%	Of actual YES, correctly predicted
Specificity	2.3%	Of actual NO, correctly predicted
Precision	75.5%	Of predicted YES, actually YES

Table 10: Classification metrics at threshold = 0.5.

Overall accuracy of 75.2% is inflated by the class imbalance and should not be taken as the primary measure of model quality. Sensitivity (2.3%) is the critical weakness: the model rarely identifies true violence cases at this threshold. Precision (75.5%) reflects that among the few cases predicted as violence, the majority are correct. Lowering the classification threshold could improve this, depending on the context in which the model is used.

6.4 Calibration Plot

Calibration analysis compares the model's predicted probabilities with the observed violence rates across decile bins. Points close to the 45-degree reference line indicate well-calibrated predictions: the model's stated confidence matches actual event rates.

The calibration plot shows an overall positive trend consistent with good calibration, though with noticeable variability in the middle probability ranges (0.15–0.35). This variability is expected given the small sample size per bin and the modest number of violence events. No systematic over- or under-estimation is detected. Larger bins provide more stable estimates.

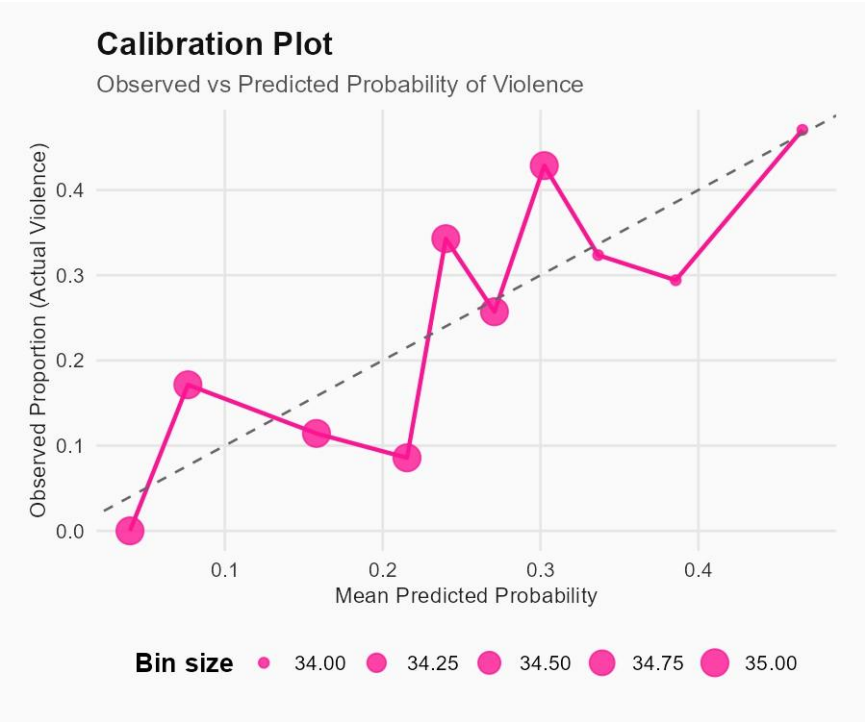

Figure 12: Calibration plot – observed vs predicted probability of violence across decile bins. The dashed 45-degree line marks perfect calibration.

7. Discussion

Interpretation of Findings

The results consistently point to socioeconomic disadvantage as a central driver of domestic violence vulnerability in this sample. Across all stages of the analysis, employment status and marital status stand out as the most influential predictors, from the initial chi-square tests and descriptive patterns through to the fully adjusted logistic regression model.

Income, while significantly associated with violence in both the Wilcoxon test and the regression model, presents a more complex picture. At first glance, it appears to be a strong predictor in the exploratory analysis, with clear differences in mean income and a higher proportion of zero-income individuals in the violence group. However, this effect becomes less straightforward once other variables are taken into account. Although log-transformed income remains statistically significant and shows a clear protective effect (OR = 0.28), its predictive strength is limited by the extreme zero-inflation in the data and the lack of variation between groups, both of which share a median of zero.

There is also some overlap between income and employment – the two variables correlate at 0.92 and income_log carries an adjusted GVIF of 11.31 – suggesting that part of income's apparent effect is being absorbed by employment status in the model. That said, the direction of the income effect is still meaningful: higher income is associated with lower odds of violence, which aligns with the idea that financial independence can reduce both reliance on an abusive partner and the practical barriers to leaving.

Education shows a non-linear pattern that deserves attention. Women with primary education have the highest violence rate (34.8%), exceeding even those with no education at all. This may reflect a transitional position where limited schooling has not yet translated into economic independence or social autonomy. Violence rates fall back to around 18% at secondary and tertiary levels, suggesting that education's protective effect strengthens meaningfully once a certain threshold is reached.

Finally, the counter-intuitive direction of the employment coefficients in the regression model (with employed women appearing at higher risk than unemployed women) is most likely a result of near-complete separation in the data. This is a known issue in logistic regression when certain categories almost perfectly predict the outcome. The extremely wide confidence intervals reinforce that these estimates are unstable and should be interpreted with caution, more as artefacts of the data than as substantive effects.

8. Limitations

• **Self-report bias and non-disclosure:** Domestic violence is a deeply sensitive and stigmatised subject. A significant concern is that some women may not have been open to disclosing their experiences, potentially under-reporting or outright denying being a victim, particularly if they felt unsafe, ashamed, or doubted that their experiences would be taken seriously. This non-disclosure could meaningfully explain the outcome distribution observed in the data, where 75.2% of participants reported no violence. The true prevalence of domestic violence in this sample may be higher than the recorded figures suggest. This is a structural limitation of survey-based research on sensitive topics and should be kept in mind when interpreting all results.

• **Small sample size:** With only 347 observations (86 in the violence group), the model works with limited statistical power. Sparse cells in some predictor combinations contribute to the near-separation instability observed in the employment coefficients.

• **Class imbalance:** The 3:1 class ratio inflates overall accuracy and severely suppresses sensitivity. Techniques such as case weights or threshold adjustment could improve the model's ability to detect violence cases in future work.

• **Cross-sectional design:** The dataset captures a single point in time and cannot establish temporal ordering or causality between predictors and violence. For example, it is unclear whether unemployment precedes or follows the onset of violence.

• **Unmeasured confounders:** Variables such as social support networks, cultural and religious factors, housing security, geographic location, and history of childhood trauma are likely relevant but not captured.

• **Income zero-inflation:** The extreme proportion of zero-income respondents limits the inferential reach of this variable. A zero-inflated model or two-part modelling approach might better characterise the income-violence relationship. Additionally, the strong correlation between income_log and employment ($r = 0.92$) means these variables are difficult to disentangle in the model, limiting the precision of their individual effect estimates.

• **Model generalisability:** The dataset origin and representativeness are not fully documented. Results may not generalise beyond the population from which the survey was drawn.

9. Conclusion

Key Takeaways & Implications

This analysis provides clear and consistent evidence that domestic violence vulnerability is strongly tied to socioeconomic circumstances. Across exploratory, inferential, and modelling stages, employment status and marital status emerge as the most powerful and reliable predictors. Income (despite its complex, zero-inflated distribution) shows a meaningful protective effect once modelled on the log scale. Education shows a nuanced, non-linear pattern that deserves further investigation.

The logistic regression model demonstrates reasonable predictive performance and good overall calibration, with no major concerns about multicollinearity. Its primary weakness – very low sensitivity for violence cases – is a well-understood consequence of class imbalance rather than a model failure, and could be addressed through resampling or threshold adjustment in future work.

01 Employment and marital status are the strongest and most consistent predictors of domestic violence risk in this sample.

02 Higher income (log-transformed) is significantly associated with reduced odds of violence, suggesting financial independence as a protective factor.

03 Most predictors show no multicollinearity concern, though income_log and employment show notable collinearity and should be interpreted with caution.

04 The model is reasonably well-calibrated despite class imbalance affecting sensitivity at the 0.5 threshold.

05 Socioeconomic interventions (employment support, income security, education access) may represent meaningful pathways to reducing domestic violence risk.

Full code, data, and visualisations available at: **github.com/cocobutterx13/Domestic-Violence-Analysis**

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